Examining How Clinical Features of Parkinson's Disease Severity Relate to Microbiome Composition and Function

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Introduction & Background

Parkinson's Disease (PD) is a common neurodegenerative disease, impacting approximately 11.8 million individuals worldwide as of 2021 (1, 2). PD is characterized by chronic and progressive motor impairment with symptoms ranging from tremors and muscular rigidity to postural instability (1). This is often accompanied by non-motor symptoms such as cognitive and gastrointestinal (GI) dysfunction (1). The clinical presentation of PD is heterogeneous among patients and can develop for decades, with pathological progression of PD often involving a slow accumulation of disability and motor impairment over time (3). As of now, no cure for PD exists and treatment primarily focuses on symptomatic care (4). Mechanisms that mediate PD susceptibility and progression are largely unknown, while identified risk factors are multifaceted and range from genetic, epigenetic, to environmental (2, 5).

Parkinson's disease can be broadly grouped into mild and severe groups. Mild PD is typically present in early stages of disease onset and can be treated with medication; however, severe PD develops during later stages of disease and is characterized by bilateral motor impairment, severe imbalance, and cognitive decline (3). Due to its diverse range of both motor and non-motor symptoms, there are many different clinical metrics used to assess the severity of PD cases. The Montreal Cognitive Assessment (MoCA) is used to detect cognitive impairment (6). Typically, MoCA scores below 26 on a 30 point scale indicate impairment (6). The Unified Parkinson's Disease Rating Scale (UPDRS) is a tool used to test both motor and non-motor symptoms of Parkinson's disease with the maximum score of 260 indicating total disability (7). The UPDRS consists of 4 different sections defined as: mood, daily living activity, motor symptoms, and other complications (7). The final metric, which is an indicator of disease severity rather than a clinical score, is disease duration. On average, longer disease duration in PD is associated with greater disease severity compared to individuals with a shorter disease duration as both motor and non-motor symptoms have accumulated over time (3).

Investigation into the factors that lead to PD neuropathogenesis may provide therapeutic avenues for not only relieving motor and non-motor symptoms in patients, but also preventing their onset. Moreover, identification of a reliable biomarker for PD diagnosis may support early intervention and monitoring of PD progression to severe disease. One emerging area of study involves the gastrointestinal system and its

connection to the brain. The gut-brain axis is defined as the bidirectional communication between the gut and central nervous system (CNS), which includes the brain and spinal cord (8). Current evidence suggests a key role for the gut-brain axis in PD. Gastrointestinal disruptions like constipation, the most common GI symptom in PD patients, and bloating often precede motor degeneration (9–11). Increased gut permeability and inflammation has also been documented in PD patients (12, 13). Further, a pathological hallmark of PD is aggregation of α-synuclein, a presynaptic terminal protein, which is found to accumulate in the gut and enteric nervous system and, in turn, propagate neuroinflammation that can result in motor degeneration (8, 10). The neuropathology by α-synuclein aggregation, however, is thought to be mediated by misfolded versions of the protein that emerge from gut inflammation induced by bacterial endotoxins and pro-inflammatory bacteria (10, 14). Taken together, this evidence points towards an alternate player in the gut that may promote PD pathogenesis – the microbiome.

The human gut microbiome is the collection of microbial genomes found in the GI tract (15). Alongside its role in metabolic and immune system regulation, the gut microbiome is a major determinant of CNS function (12, 15). Several interactions between the gut microbiome and CNS have been established in neurological diseases like Alzheimer's and multiple sclerosis. Research is increasingly examining how the gut microbiome may function in PD pathogenesis via the gut-brain axis (8). In addition to the above study highlighting how bacteria-mediated gut inflammation induces misfolded α-synuclein, specific genera of the gut microbiome in PD patients like *Dubiosella* are shown to cause metabolic dysfunction and promote α-synuclein aggregation in microglia, which are resident immune cells of the brain responsible for α-synuclein degradation (16). Moreover, gut microbiome dysbiosis and dominance of pro-inflammatory genera in PD patients may produce clinical observations like increased gut permeability that harbors neuroinflammation via the gut-brain axis (9, 17, 18). Overall, there is a clear link between the gut microbiome and PD, prompting investigation into how gut microbiome alterations influence PD pathogenesis.

Alpha diversity is informative of the overall compositional space of the microbiome in each patient, but it cannot specify taxa differences in the community that may be functional in PD pathogenesis (19). Numerous studies have shown there are no significant differences in alpha diversity in healthy controls (HC) compared to PD patients, yet there are differences in gut microbiome taxa seen across different cases of PD severity while controlling for patient age and sex, which are variables associated with altered microbiome composition (5, 20, 21). There is mounting evidence of a pro-inflammatory microbiota in PD, including increased levels of inflammation-associated microbial genera that may contribute to neuroinflammation (9). Further, a longitudinal study by Lubomski et al. has shown that faster-progressing

PD, as determined by symptom accumulation, may be associated with taxa differences such as decreased abundance of *Barnesiella*, a bacterial family shown to prevent pathogenic enterococci from colonizing the gut (21). However, little remains known about how PD severity may potentially correlate to altered gut microbiome composition.

In this project, we aim to use the three measures of disease severity outlined above – MoCA, UPDRS, disease duration – to explore the possibility of a correlation between disease severity metrics of PD and the composition and function of the gut microbiome. First, preliminary analyses will explore correlations between the alpha diversity of PD patients and their clinical measures , analyzing each of the three measures independently. Next, we aim to bin patients into PD severity categories (healthy, mild PD, severe PD) to examine the combined effects of the three clinical measures on the gut microbiome composition and function. Organization of severity categories will be accomplished using preexisting thresholds for severity outlined by the MoCA and UPDRS as well as clinical symptoms accumulating over prolonged disease duration. . In conjunction, investigation of the preliminary alpha diversity analysis may contribute to the categorization efforts if a significant shift in microbial composition is observed over specific scoring thresholds. Using the binned categories, we aim to perform a beta diversity analysis to identify microbial taxa and their respective metabolic functions as potential biomarkers defined by PD severity groups. Finally, we aim to complete an in-depth functional analysis of the significant taxa to explore the implications of PD severity on patient gut microbiome health and function. Overall, exploring changes in gut microbiome composition and function across PD severity metrics could provide insights on therapeutic targets for PD and potential biomarkers of PD onset and progression.

Research Question/Hypothesis

Research Question: Is Parkinson's disease severity, defined by cognitive decline (MoCA), motor impairment (UPDRS), and disease duration, associated with changes in gut microbiome composition and function?

Hypothesis: We hypothesize that increased Parkinson's disease severity metrics (lower MoCA scores, higher UPDRS scores, and longer disease duration) are associated with depletion of beneficial short-chain fatty acid (SCFA)-producing taxa and enrichment of pro-inflammatory taxa.

This study aims to investigate how PD severity, defined by a patient's MoCA score, UPDRS score, and disease duration, impacts the composition and function of the gut microbiome using data from a cross-sectional cohort study of 300 human participants (22). Based on previous literature, we hypothesize

that increased disease severity (lower MoCA scores, higher UPDRS scores, and longer disease duration) will lead to pathogenic changes such as decreases in alpha diversity, beta diversity, and differential abundance of the gut microbial community that play a functional role in PD neuroinflammation. Specifically, we hypothesize that patients with more severe disease scores will display a greater reduction in beneficial SCFA-producing taxa and an enrichment of pro-inflammatory or opportunistic taxa.

Prior literature has documented gut dysbiosis in PD patients compared to healthy control patients most commonly defined by a depletion in *Prevotellaceae* and *Faecalibacterium* (23, 24). *Prevotellaceae* are anaerobic SCFA-producing bacteria which are known to be beneficial in producing SCFAs during fermentation of dietary fiber. SCFAs are known to play a key role in the regulation of energy metabolism and supply and are also instrumental in maintaining homeostasis within the intestinal tract of the gut (25). Decreases in *Faecalibacterium* have also been observed across PD patients where its reduced abundance is associated with inflammatory conditions in the gut due to interruptions in SCFA-production necessary in regulating inflammation and maintaining the gut epithelial barrier (23, 24, 26–28). Decreases in the abundance of these taxa suggest imbalances in PD patients' gut digestion.

Research investigating gut bacterial profiles in PD patients have observed an increase in *Bifidobacterium* and *Akkermansia* (24). Although these genera are typically present in a healthy gut environment as beneficial taxa, their enrichment coupled with the depletion of SCFA-producing bacteria have been linked to dysbiosis resulting in a pro-inflammatory metabolic state in PD patients (12). Intestinal inflammation has recurrently been documented as a symptom of PD alongside mucosal barrier disruption which has been hypothesized to propagate α-synuclein pathology via the gut-brain axis (13, 17, 29). Studies tracking fecal biomarkers have suggested that this pro-inflammatory metabolism may be linked to bacterial gut dysbiosis (13, 17). Therefore, investigating the change in diversity throughout PD progression is crucial for treatment development as probiotics, prebiotics and fecal microbiota transplantation are being further studied as potential interventions to improve PD patients' gut health and quality of life (30). Our study aims to determine whether severe motor impairment, cognitive dysfunction, and disease duration have differing impacts on gut microbiome modulation to improve treatments for patients experiencing gut dysbiosis.

Experimental Aims & Rationale

Aim 1: To determine whether gut microbial diversity is significantly correlated with each clinical severity measure of PD (MoCA score, UPDRS score, disease duration).

Parkinson's disease is characterized by many gastrointestinal comorbidities such as constipation, reduced colonic transit time and altered gut microbiota (11, 18). How changes in microbiota composition relate to function of the GI tract in Parkinson's disease remain largely unknown. Cirstea et al. collected data from 300 participants, including PD patients and HC in a cross-sectional cohort study (22). The data collected focused on motor and non-motor PD symptoms, medications, diet, and different demographic factors. The results showed significant microbiota taxonomic differences in PD patients compared to HC patients, with PD patients functionally characterised by reduced carbohydrate fermentation and synthesis capacity of the SCFA, butyrate. However, these results are based only on the presence or absence of PD and lack comparison between Parkinson's disease severity and measures of microbial diversity.

Aim 1.1: Alpha diversity and PD severity correlation

After initial data processing in QIIME2, we will load the data into R to complete data processing and finally create a new Phyloseq object (31). Using the picante R package, we will calculate Faith's phylogenetic diversity for the alpha diversity of PD patient samples from the new Phyloseq object and add that into the metadata table as a new column (32). Faith's phylogenetic diversity takes into account community richness and phylogenetic relatedness between features (32). Alpha diversity analysis will be conducted independently for each PD severity measure – MoCA, UPDRS, disease duration – to assess if there is a significant difference in microbial richness and evenness within each of the different clinimetrics of PD severity. Significance will be tested with Spearman's statistical test to assess correlation strength. If a significant correlation is observed between alpha diversity and each measure of disease severity, we expect it to be negative as severity increases (lower MoCA scores, higher UPDRS scores, and longer disease duration) since decreased microbial diversity is associated with many chronic, inflammatory, and metabolic diseases including other neurodegenerative diseases (15).

Aim 1.2: PD severity metric binning

Next, we aim to organize patient samples in the dataset into PD severity categories (healthy, mild PD and severe PD) for investigation into how gut microbiome composition differs across binned PD severity measures. If we observe a correlation between any of the PD severity measures, this can inform categorizations and allow us to transform the continuous MoCA and UPDRS scores and disease duration into discrete categories for downstream beta diversity analysis. For example, if a significantly dramatic shift in alpha diversity is observed at a specific

point of disease duration, it can be utilized as a binning threshold as it may indicate a microbiome alteration that is functional in PD pathogenesis, enabling insight into a PD therapeutic target. If no significant alpha diversity correlation is observed, the different severity measures will be binned based on preexisting thresholds. Both the MoCA and UPDRS metrics have documented thresholds for mild, moderate and severe cognitive impairment and motor evaluation, respectively (6, 7).

Aim 2: To evaluate if the overall microbial community structure differs significantly between healthy controls and PD patients across different levels of disease severity. This aim will focus on determining if the severity bins established in Aim 1 belong to distinct microbial clusters through beta diversity analysis.

Aim 2.1: Weighted UniFrac analysis

Using the categorical bins (healthy, mild PD, severe PD), we will perform beta diversity analysis using Weighted UniFrac. This method will account for both phylogenetic closeness of taxa and their relative abundances to investigate which pro-inflammatory taxa may dominate the gut as PD severity increases (33). To validate whether the disease severity is the main driver of these shifts, we will then perform a PERMANOVA test using the vegan package in R (34). We expect to observe clustering corresponding to severity levels which would suggest the gut environment undergoes transition as motor and cognitive function declines.

Aim 3: To identify specific microbial taxa and metabolic functions that define different PD severity groups.

While alpha and beta diversity metrics from Aims 1 and 2 will provide a general overview of community shifts, we aim to further identify the specific taxa and metabolic functions that characterize different stages of Parkinson's disease severity. This aim focuses on narrowing the scope to specific microbial taxa that characterize clinical decline. By isolating them, we can better understand how the gut-brain axis may host beneficial or pathogenic microbes during different stages of neurodegeneration, thus allowing us to find biomarkers associated with PD pathogenesis.

Aim 3.1: Core microbiome

We will identify the core microbiome to determine which taxa are consistently present in the sample within a group despite individual variations. In R we will use the microbiome package's *core_members()* function to identify these stable community members for our healthy, mild PD, and

severe PD bins (35). By comparing the core microbiome of healthy controls to patients with differing levels of clinical severity, we can identify which beneficial species are lost or which potentially harmful species are gained during disease progression. This will inform us if PD progression is determined by the stability of pathogenic taxa or instability of healthy ones.

Aim 3.2: Indicator taxa analysis

Using the indicspecies package in R, we will identify taxa that are highly specific and prevalent in each severity group (36). This will identify key pro-inflammatory species that characterize the transition from healthy to severe PD bins.

Aim 3.3: DESeq2 analysis

We will identify species that are differentially abundant in the gut microbiome environment of the binned groups from Aim 1.2. This will be completed using the DESeq2 package in R to conduct a DESeq2 analysis across the different severity groups observed (37). Differentially abundant bacteria present at a $p < 0.05$ will be considered statistically significant and visualized using a bar plot and heat map using the *ggplot()* and *pathway_heatmap()* functions, respectively (38, 39). Following identification, a literature review examining metabolic functions within the gut microbiome will be conducted to elucidate the role of differentially expressed taxa within each clinical severity group (healthy, mild PD, and severe PD).

Aim 3.4: Random Forest model

To determine which microbial taxa best define the various PD severity groups, Random Forest will be used to rank taxa based on their importance in predicting severity level (healthy, mild PD, severe PD). The scores generated for the microbial taxa will allow identification of the most influential microbes distinguishing PD severity.

Aim 3.5: Use PICRUSt2 to identify the metabolic pathways that are enriched or depleted

We will carry the differentially abundant taxa identified across severity groups into functional analysis using Phylogenetic Investigations of Communities by Reconstruction of Unobserved States 2 (PICRUSt2) to predict metabolic pathway activity associated with each PD severity group (39). Predicted pathways will be mapped to KEGG for comparison to identify functional shifts associated with PD progression.

Proposed Approach

Data Processing	
1	Import raw V4 16S rRNA sequences from manifest.tsv file into QIIME2 to create demux.qzv (31)
2	Use DADA2 to denoise, trim and cluster the sequences to create rep-seqs.qza and table.qza (40)
3	Use gg-2024-515F-806R-classifier.qza to assign taxonomy to the reads, generating taxonomy.qza
4	Filter the dataset to exclude mitochondrial taxonomy assignments, generating table-no-mitochondria.qza
5	Generate a phylogenetic tree file, rooted-tree.qza, using rep-seqs.qza
6	Use the rooted-tree.qza and table-no-mitochondria.qza to generate alpha-rarefaction.qzv
7	Export all files from QIIME2 and use QIIME2view
8	Determine the plateau on observed_features to validate the rarefaction number that will be used for downstream analysis

Aim 1: Determine if there is a correlation between gut microbial diversity and clinimetrics of PD severity (MoCA, UPDRS score, disease duration).	
1	Load phyloseq, ape, tidyverse, and picante packages and rarefied data into R (32, 41, 42, 43)
2	Generate a new phyloseq object, combining feature table, taxonomy, phylogeny, and metadata.
3	Subset the new phyloseq object to include only PD patients using *subset()*.
4	Calculate Faith's phylogenetic diversity for the alpha diversity of PD patient samples from the Phyloseq object and combine into a metadata table.

5	Use *ggplot()* to plot alpha diversity of PD patient samples against MoCA score ('moca_total'), UPDRS total ('updrstotal'), and disease duration ('Disease_dur') respectively in a scatterplot using *geom_point()*.
6	Perform statistical analysis using Spearman's test to assess correlation strength between alpha diversity of PD patients and severity metrics.

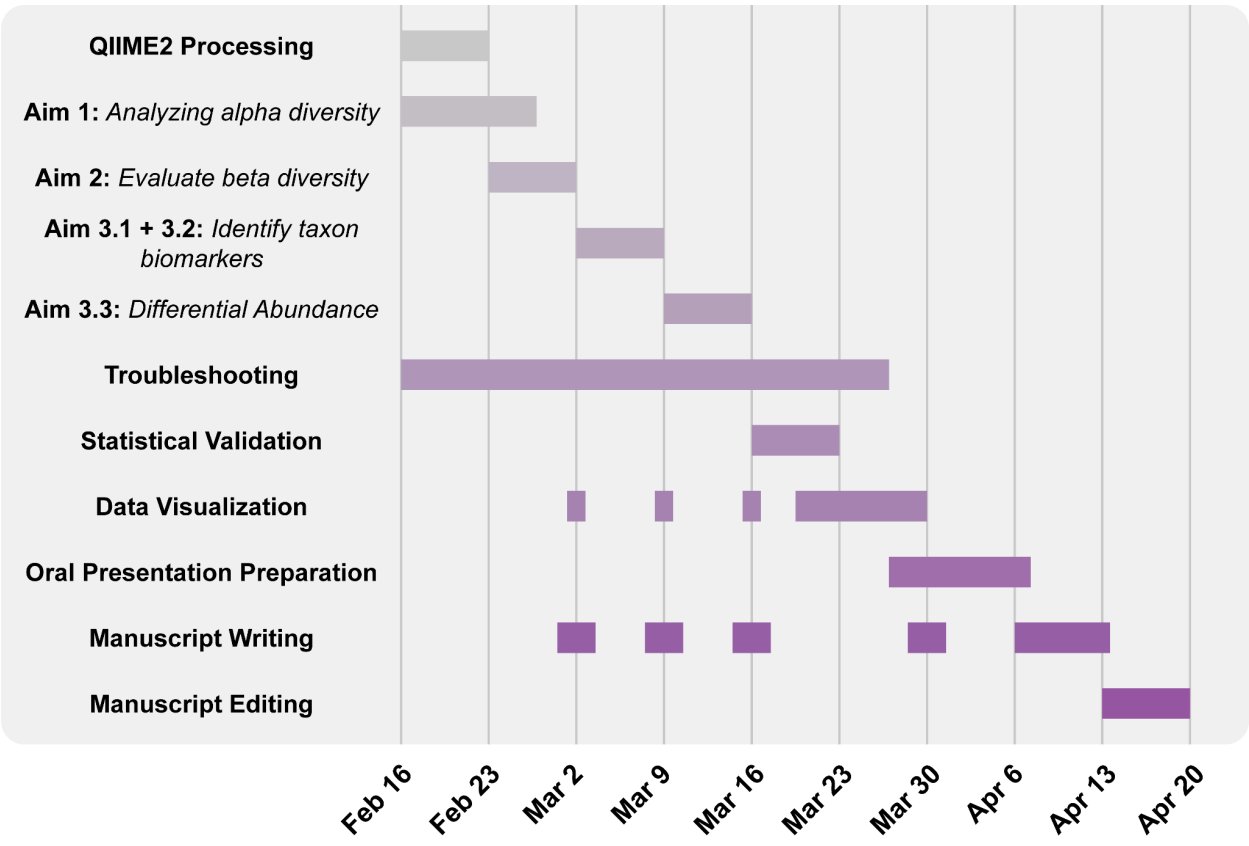
Aim 2: Evaluate if overall microbial community structure differs significantly between healthy controls and PD patients across different levels of severity using beta diversity.	
1	Load phyloseq, vegan, ggplot2, microbiome and tidyverse package (34, 35, 38, 41, 43)
2	Load phyloseq object generated in Aim 1 and perform beta diversity using weighted unifrac
3	Subset phyloseq object to include only the "Healthy" group using *subset()*
4	Perform beta diversity on the subsetted object using Weighted UniFrac
5	Subset phyloseq object to include only the "Mild PD" group using *subset()*
6	Perform beta diversity on the subsetted object using Weighted UniFrac
7	Subset phyloseq object to include only the "Severe PD" group using *subset()*
8	Perform beta diversity on the subsetted object using Weighted UniFrac
9	Perform Faith's phylogenetic alpha diversity again but using the binned categories.
10	For all groups, perform statistical analysis using PERMANOVA test in R (44)

Aim 3: To identify specific microbial taxa that define PD severity groups.

1	Load microbiome, indicspecies, DESeq2, and ggVennDiagram packages in R (35, 36, 37, 39, 45)
2	Use binned phyloseq object using the clinical severity categories (healthy, mild PD, and severe PD) established in Aim 1.2 (34)
3	Use *core_members()* function from the microbiome package to identify the core microbiome, taxa present in the samples for each severity group. (35)
4	Generate a Venn Diagram (*ggVennDiagram()*) to visualize unique and shared taxa across healthy, mild PD, and severe PD groups (45)
5	Perform indicator species analysis using *multipatt()* from indicspecies package to identify taxa with high specificity to each level of severity. (36)
6	Identify significant indicator taxa by evaluating Indicator Values and p-values
7	Conduct differential abundance analysis using DESeq2 to calculate the log2-fold changes of taxa between different severity bins (comparing to the healthy control) and generate heat map and volcano plots. (37)
8	Use *ggplot()* to generate various plots to visualize microbial shifts.
9	Use the feature table of microbial taxa abundances and corresponding PD severity metadata from the phyloseq object for machine learning analysis
10	Train a Random Forest classification model using taxa abundances as predictor variables and PD severity category (healthy, mild PD, severe PD) as the response variable to determine taxa that best predict disease severity.
11	Extract feature importance scores from the Random Forest model and rank taxa according to their contribution to classification performance.

12	Use the significant indicator taxa and top-ranked Random Forest taxa for functional prediction using PICRUSt2 to infer gene family abundances (39),
13	Map predicted gene families to metabolic pathways using the KEGG and compare pathway enrichment across healthy, mild PD, and severe PD groups to identify functional shifts associated with PD progression.

Weekly Timeframe



Week	Objective
Completed	**QIIME2 data processing:** - Import raw sequence > DADA2 denoising, taxonomy assignment, truncating reads to 220 nucleotides, rarefaction depth at 8000.

Feb 16 - 20	**Aim 1:** Analyzing alpha diversity - Generate phyloseq objects - Calculate Faith's phylogenetic diversity to assess correlation with MoCA, UPDRS, and disease duration via Spearman's test
Feb 23 - 27	**Aim 2:** Evaluate beta diversity - Use Weighted UniFrac metrics and PERMANOVA to determine if community structure differs significantly across severity bins
Mar 2 - 6	**Aim 3.1 + 3.2:** Identify taxon biomarkers - Run core microbiome and indicator taxa analysis to isolate species gaining or losing prevalence as PD progresses
Mar 9 - 13	**Aim 3.3:** Differential Abundance - Conduct DESeq2 analysis to identify specific genes with significant log2-fold changes between severity groups
Mar 16 - 20	**Statistical Validation + Troubleshooting:** - Re-running any analysis to confirm results
Mar 23 - 27	**Data Visualization + Manuscript:** - Finalize figures (PCoA plots, heatmaps, scatter plots, and etc) using ggplot2 - Draft results and discussion sections
Mar 30 - Apr 3	**Oral Presentation Preparation:** - Finalize final presentation slides

<u>Dataset Overview</u>

This study is using data sourced from Cirstea et al.'s research *"Microbiota composition and metabolism are associated with gut function in Parkinson's disease",* published in Movement Disorders (22). The dataset contains single-end sequenced V4 regions of 16S rRNA from 300 fecal samples, representing a cohort of individuals categorized into those with PD and healthy controls. The participants range in age from 42 to 80 years old with a sex distribution of 128 females and 172 males. The data set also provided our metrics of interest with MoCA scores ranging from 17 to 31, the total UPDRS score ranging from 18

and 120, and disease duration spanning 1 to 20 years. Beyond these, the metadata contains comprehensive physiological and clinical details, such as Body Mass Index (BMI) ranging from 16.7 to 58, depression scores, and gastrointestinal symptoms including constipation severity and IBD status.

The DADA2 tool, as mentioned in Callahan et al.'s work *"DADA2: High resolution sample inference from Illumina amplicon"*, was used to denoise in QIIME2 (v2025.04) (40). Sample size and sequencing depth were quantified before and after denoising (Table 1).

Table 1: Sample size and sequencing depth

	Before Denoising	**After Denoising**
Total Sample Size	300	300
Sequencing Depth	3,673–29,952	3,465–24,486

Read length and quality score were determined after demultiplexing and viewing the file using QIIME2view (31). Demultiplexing the dataset showed all read lengths to be uniformly 251 nucleotides (nts). Based on the quality plot, we decided to truncate the reads to 220 nts since that is when the quality begins to drop below a score of 30 (Figure 1). A quality score ≥30 indicates an accuracy >99.9%.

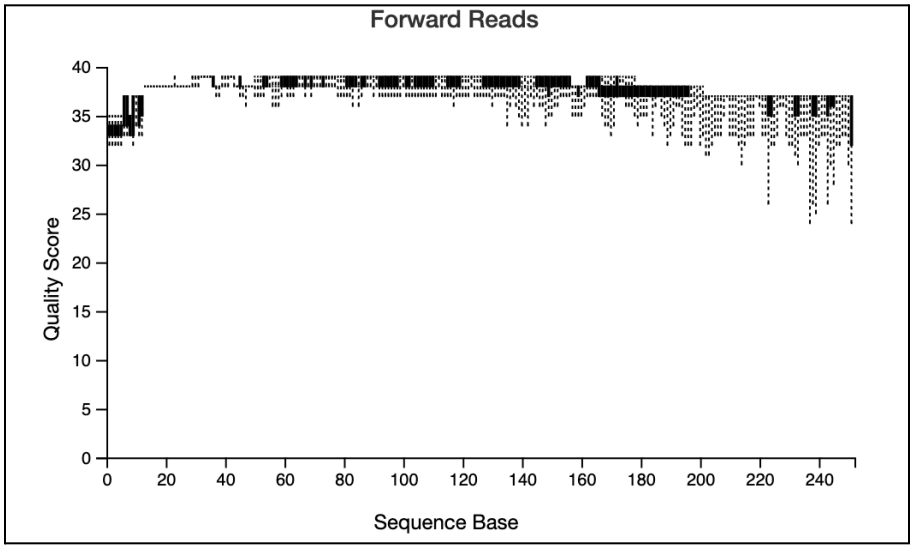


Figure 1: Read Quality Score Plot. Quality score for each base was generated with QIIME2 random sampling 10,000 out of 4,481,306 sequences without replacement.

Reads that were identified to be mitochondria or chloroplast (non-bacterial) sequences, were filtered out in QIIME2 (31). Further filtering of the dataset will be done in RStudio depending on the severity parameter (46). Our analysis focuses on disease severity, measured by disease duration (disease_dur), cognitive impairment (moca_score) and motor impairment (updrs_total) (22). The majority of analyses will require only patient samples that have Parkinson's Disease (PD). Data regarding disease duration and motor impairment is only available for samples with PD, no data is available for the healthy control samples. Only cognitive impairment data is available for both PD and healthy control samples. Possible confounding variables, including age, sex, BMI, and education (confounder for cognitive impairment) is available for all samples while the age onset with Parkinson's (confounder for disease duration) is only available for samples with PD (7).

Rarefaction depth was determined in QIIME2view with the table.qzv and alpha-rarefaction.qzv files (31). We generated the alpha-rarefaction curve using a depth of 25,000 since our maximum frequency per sample is 24,486 (Figure 2). The metadata was taken from the column indicating whether or not the patient sample has Parkinson's Disease (PD). A sequencing depth of 8,000 was chosen since that is where the curve begins to plateau (Figure 2). Using the QIIME2view interactive sample plot, 2,152,000 (56.76%) features in 269 (89.67%) samples are retained at a sampling depth of 8,000 (31). There would be 18 samples lost from the 197 samples that have PD and there would be 13 samples lost from the 103 healthy control samples.

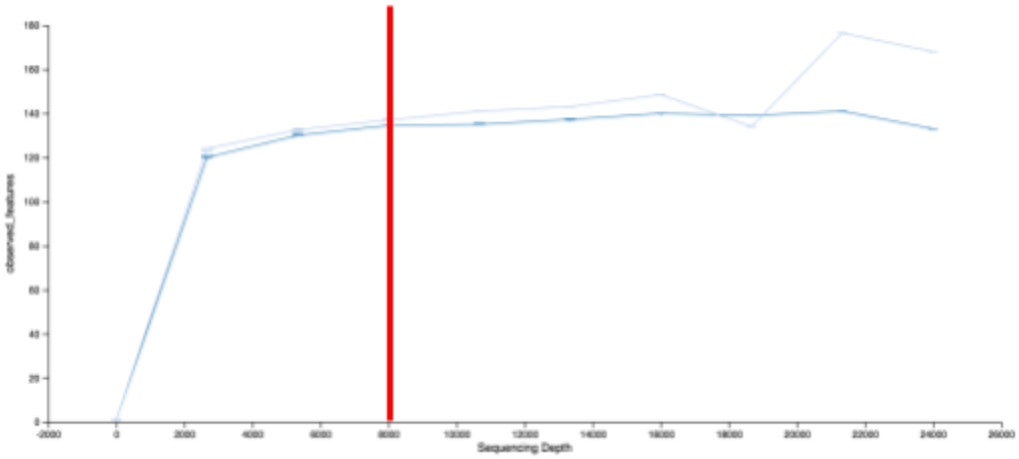


Figure 2: Alpha rarefaction plot generated using QIIME2view. Dark blue represents the control samples. Light blue represents the samples with Parkinson's Disease. Sample metadata was taken from the 'Disease' column. The sampling depth is set to 8,000 as indicated by the red line.

Team Contributions

Teammate	Proposal Contributions	Analysis
Narjis Alhusseini	Hypothesis/Prediction, Aim 2, Aim 3.3, References, Zotero setup	Aim 2/3, data visualization
Jaclyn Armstrong	Introduction & background, Aim 1, References	Aim 3.3, statistical validation
Ella Buric	Introduction and background, Aim 1, Hypothesis/Prediction, References	Aim 1
Matthew Chan	Aim 3.1, 3.2, Timeline (Gantt Chart), References	Aim 3
Mikaela Pirillo	Research question, Dataset overview, References	QIIME2 processing, Aim 2

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